



0082_Akashdeep Pal <akashdeep82.be22@chitkara.edu.in>

IEEE Chandigarh Subsection International Conference : Submission (1889) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org>

Mon, May 11, 2026 at 6:29 PM

To: akashdeep82.be22@chitkara.edu.in

Hello,

The following submission has been created.

Track Name: Security & Privacy

Paper ID: 1889

Paper Title: The Rising Threat of Ransomware: Trends, Impact, and Prevention Strategies

Abstract:

There is little doubt that ransomware is now considered the costliest cybercrime threat that causes immense disruptions to healthcare, critical infrastructure, government entities, and commercial firms. This article conducts an extensive empirical study of ransomware development during the period 2015-2025, including notable ransomware families, Ransomware-as-a-Service (RaaS), and double extortion tactics that have profoundly changed the threat environment. In this paper, we assess vectors of attacks, propagation techniques, encryption algorithms, and payment schemes used by key ransomware players. Based on a database comprising 4,700+ ransomware attacks recorded, we study temporal trends in ransom amounts, affected industries, and restoration performance. In addition, we develop the innovative concept of the Ransomware Resilience Framework (RRF) that operates on zero-trust, behavioral AI, immutability backups, and incident response. The performance of the RRF is estimated, suggesting a projected 94% decrease in encryption attack incidences and 78% improvement in mean time to restore..

Created on: Mon, 11 May 2026 12:59:29 GMT

Last Modified: Mon, 11 May 2026 12:59:29 GMT

Authors:

- akashdeep82.be22@chitkara.edu.in (Primary)
- gaurav1571.be22@chitkara.edu.in
- shivam2315.be22@chitkara.edu.in

Secondary Subject Areas: Not Entered

Submission Files:

G4_The_rising_threat_of_ransomware_RP_new.docx (44 Kb, Mon, 11 May 2026 12:57:45 GMT)

Submission Questions Response:

1. Corresponding author Email ID

gaurav1571.be22@chitkara.edu.inshivam2315.be22@chitkara.edu.in

2. Contact number with country code

+919696017815

+917807443716

3. Author's Information

Agreement accepted

4. Co-author's Consent

Agreement accepted

5. Certificate of Originality

Agreement accepted

6. Author's Consent

Agreement accepted

7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

Agreement accepted

Thanks,
CMT team.

Please do not reply to this email as it was generated from an email account that is not monitored.

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our [Privacy Statement](#).

Microsoft Corporation
One [Microsoft Way](#)
[Redmond, WA 98052](#)